Week 6 Part 1

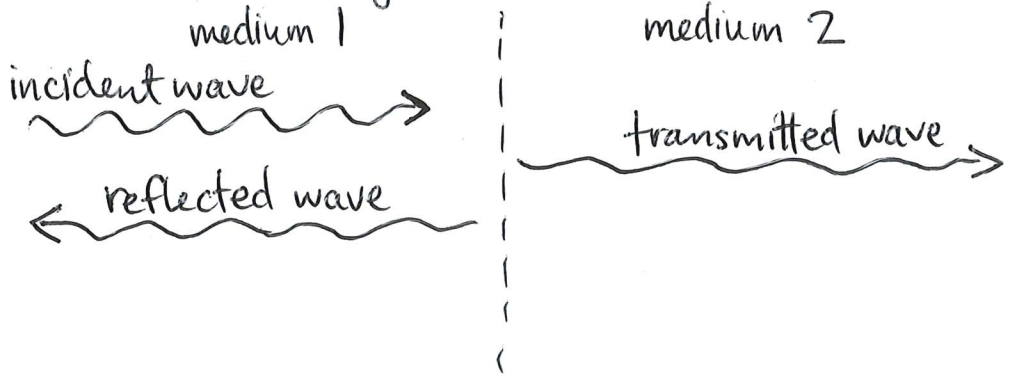
PHYS 369

So far we have been looking at how to describe waves, their properties, and the media they travel through. As a quick reminder, we have examined:

- simple harmonic motion to describe particle motion in media
- the wave equation; wave speed, angular frequency, wavenumber
- complex representation of waves for adding waves together
- energy transported by waves, power, intensity
- acoustic impedance as measure of media resisting passage of waves
- sound level, geometric spreading, attenuation

All of this has been to lay the foundation for the more complicated topics coming next (complicated in terms of wave dynamics).

Today we will be looking at reflection and transmission of waves as they pass from one medium to another, which will require us to use many of the ideas we have been developing. Consider the following scenario:



For convenience we can choose $\phi_i = 0$ of the incident wave so that we are measuring the phase constants ϕ_r and ϕ_t with respect to the incident wave. This allows us to write

$$\tilde{y}_i(x,t) = A_i e^{j(\omega_i t - k_i x)} \qquad (A_i \text{ is real})$$

$$\tilde{y}_r(x,t) = \tilde{A}_r e^{j(\omega_r t + k_r x)} \qquad (\tilde{A}_r = A_r e^{j\phi_r})$$

$$\tilde{y}_t(x,t) = \tilde{A}_t e^{j(\omega_t t - k_t x)} \qquad (\tilde{A}_t = A_t e^{j\phi_t})$$

$\Rightarrow$ Top Hat question

What else can we say about the 3 waves? To think about this physically, consider a wave travelling down a string made of 2 strings joined together:



When the pulse hits the junction b/w the two pieces, there will be a reflected wave and transmitted wave. The wave equation is a 2nd order ODE, so there cannot be any discontinuities in the rope and there cannot be any kinks that form (the latter would imply infinite acceleration at the kink)

The first condition requires

$$\boxed{(1) \quad A_i + \tilde{A}_r = \tilde{A}_t} \qquad \text{at } x = 0$$

* note: the condition is <u>not</u> that $|A_i| + |A_r| = |A_t|$

The second condition requires

$$(2)\quad \frac{d}{dx}\tilde{y}_i + \frac{d}{dx}\tilde{y}_r = \frac{d}{dx}\tilde{y}_t \qquad \text{at } x=0$$

since $\omega_i = \omega_r = \omega_t = \omega$, $k_i = k_r = k_1$ and $k_t = k_2$

$$\tilde{y}_i = A_i e^{j(\omega t - k_1 x)}$$
$$\tilde{y}_r = \tilde{A}_r e^{j(\omega t + k_1 x)}$$
$$\tilde{y}_t = \tilde{A}_t e^{j(\omega t - k_2 x)}$$

Therefore condition (2) becomes:

$$-jk_1 A_i + jk_1 \tilde{A}_r = -jk_2 \tilde{A}_t$$

or

$$k_1 A_i - k_1 \tilde{A}_r = k_2 \tilde{A}_t$$

Note that $Z \propto \frac{1}{v}$ and $k_1 = \frac{\omega}{v_1}$, $k_2 = \frac{\omega}{v_2}$ so $k = CZ$

for any medium with C being a constant that does not change at the boundary. We can therefore re-express condition (2) as:

$$\boxed{(2)\quad Z_1 A_i - Z_1 \tilde{A}_r = Z_2 \tilde{A}_t}$$

5 minute break

Now that we have the boundary conditions, let's use them. First we define the reflection and transmission coefficients:

$$\mathcal{R} = \frac{\tilde{A}_r}{A_i} \quad \text{and} \quad \mathcal{T} = \frac{\tilde{A}_t}{A_i} \qquad * \text{both are complex valued} *$$

Conditions (1) and (2) then yield:

$$(1)\quad 1 + \mathcal{R} = \mathcal{T} \qquad \text{and} \qquad (2)\quad 1 - \mathcal{R} = \frac{Z_2}{Z_1}\mathcal{T}$$

This is a system of linear equations for $\mathcal{R}$ and $\mathcal{T}$ so we can solve them using standard techniques:

$$(1) + (2): \quad 1 + \cancel{\mathcal{R}} + (1 - \cancel{\mathcal{R}}) = \mathcal{T} + \left(\frac{Z_2}{Z_1}\mathcal{T}\right)$$

$$\Rightarrow \boxed{\mathcal{T} = \frac{2Z_1}{Z_1 + Z_2}}$$

Sub this $\mathcal{T}$ into (1): $\mathcal{R} = \frac{2Z_1}{Z_1 + Z_2} - 1$

$$\Rightarrow \boxed{\mathcal{R} = \frac{Z_1 - Z_2}{Z_1 + Z_2}}$$

Immediately notice: $\mathcal{T}$ and $\mathcal{R}$ are complex valued by their definitions, but the impedances are real, so $\mathcal{T}$ and $\mathcal{R}$ must also be real. This seems like a contradiction but all it means is that ϕ_r and ϕ_t are heavily constrained to be either 0 or π by the boundary conditions. That is to say both the reflected and transmitted waves must be <u>in phase</u> or <u>π rad out of phase</u> with the incident wave. No other options are possible.

Let's look at some specific cases:

1) $Z_1 = Z_2$

The trivial case would be when the medium is exactly the same on either side of the interface. However, this does not need to be the case because what is required is the acoustical impedance is equal on both sides.

Example: ultrasound probe.

2) $Z_2 \gg Z_1$

This would correspond to waves reflecting off a very stiff boundary. Examples: rope anchored to a wall, sound waves hitting a rigid surface, etc.

Here we would have: $$T = \lim_{\frac{Z_2}{Z_1} \to \infty} \frac{2Z_1}{Z_1 + Z_2} = \lim_{\frac{Z_2}{Z_1} \to \infty} \frac{2}{1 + Z_2/Z_1} = 0$$

$$R = \lim_{\frac{Z_2}{Z_1} \to \infty} \frac{Z_1 - Z_2}{Z_1 + Z_2} = \lim_{\frac{Z_2}{Z_1} \to \infty} \frac{1 - Z_2/Z_1}{1 + Z_2/Z_1} = -1$$

There is zero transmission and $R = \frac{\tilde{A}_r}{A_i} = \frac{A_r}{A_i} e^{i\phi_r} = -1$

so $A_r = A_i$ (same amplitude) and $\phi_r = \pi$ (180° out of phase).

The reflected wave is equal and inverted:



3) $Z_1 \gg Z_2$

This is like a rope with the end free instead of anchored to the wall.

$$R = \lim_{\frac{Z_1}{Z_2}\to\infty} \frac{Z_1 - Z_2}{Z_1 + Z_2} = \lim_{\frac{Z_1}{Z_2}\to\infty} \frac{Z_1/Z_2 - 1}{Z_1/Z_2 + 1} = 1$$

$$\tau = \lim_{\frac{Z_1}{Z_2}\to\infty} \frac{2Z_1}{Z_1 + Z_2} = \lim_{\frac{Z_1}{Z_2}\to\infty} \frac{2\,Z_1/Z_2}{Z_1/Z_2 + 1} = 2$$

The wave is reflected in phase with the incident wave ($R=1$) but how can it be that $\tau = 2$? If we look at the power reflected and transmitted:

$P = \frac{1}{2} Z \omega^2 A^2$ so we define

$$R_e \equiv \frac{P_r}{P_i} = \frac{\frac{1}{2} Z_1 \omega^2 |\tilde{A}_r|^2}{\frac{1}{2} Z_1 \omega^2 A_i^2} = \frac{|\tilde{A}_r|^2}{A_i^2} = R^2$$

$$\tau_e = \frac{P_t}{P_i} = \frac{\frac{1}{2} Z_2 \omega^2 |\tilde{A}_t|^2}{\frac{1}{2} Z_1 \omega^2 A_i^2} = \frac{Z_2}{Z_1} \frac{|\tilde{A}_t|^2}{A_i^2} = \frac{Z_2}{Z_1} \tau^2$$

Clearly $R_e = 1$ (all power is reflected) and $\tau_e = 0$ (no power is transmitted). This is similar to elastic collisions where an object M hits an object $m \ll M$: the massive object is unaffected and travels at same speed v while the small object travels at speed 2v but all the kinetic energy is still carried by M since it's so much larger.